A Design-choice quality cost (95% CI)

tail32 subspace (E133)
tail32 bit-identical (E129)
dyn gate ε=.05 (E137)
dyn gate ε=.10 (E137)
static gate (E129)
sink+SWA only (E135a)

−30 −25 −20 −15 −10 −5 0 5

AVG5 paired delta vs full anchor (points)

B Representation arms (L1/L2 scoring dims)

steady-state GPU time (ms/step)

0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6

1.3502 1.3598 1.4634

4x coarse compute saved
at +0.7% latency
(sub+sub vs full+sub)

full+sub sub+sub sub+full

C K2 budget: decode cliff vs prefill recovery

task score

10 20 30 40 50 60 70

dense hotpot 59.4

dense musique 34.3

decode hotpotqa
decode musique
prefill hotpotqa (grp64)
prefill musique (grp64)

256 384 512 768 1024 1536 2048

5×10^{2} 6×10^{2} 2×10^{3}

K2 token budget per query

D Middle selection value (M0 vs M1)

task score

0 10 20 30 40 50 60 70

FullKV (M0)
sink+SWA only (M1, 256 tok)

-35 -30 -22 -11 -12

qasper hotpotqa musique gov_rep repo